

Simple E-commerce Product Browser — Product Requirements Document

1. Executive Summary

This document outlines the requirements for the Simple E-commerce Product Browser, a foundational web application designed to provide a straightforward and intuitive platform for online shoppers. Its primary purpose is to enable users to effortlessly browse a catalog of products, view detailed information for each item, and compile a list of desired products within a shopping cart. This initial release focuses on validating the core user journey of product discovery and selection, addressing the need for a frictionless way for shoppers to explore and curate potential purchases without immediate commitment, thereby streamlining the initial selection process for future e-commerce interactions.

2. Problem Statement

Online shoppers today often face friction when trying to discover and compile a list of desired products without the immediate pressure of a full purchase commitment. The current state for many users involves disjointed browsing experiences, difficulty in temporarily saving items of interest, or the need to proceed directly to checkout to simply "hold" items. This leads to a suboptimal user experience, potential abandonment of browsing sessions, and missed opportunities for retailers to capture initial user interest.

The specific problem we are solving is the lack of a straightforward and intuitive platform for users to discover products and compile a list of desired items without immediate commitment to purchase. This problem is experienced by online shoppers who are in the discovery or consideration phase, and by individuals seeking to browse products for future reference or comparison. The severity of this problem manifests as user frustration, reduced engagement with product catalogs, and an inability to easily create a personal shopping list within a digital storefront.

The consequences of not solving this problem include a higher bounce rate on product pages, lower conversion rates in subsequent purchase stages (as initial interest isn't effectively captured), and a diminished overall user experience. The current state is a fragmented browsing journey where saving items is cumbersome or

requires immediate purchase intent. The desired state is a seamless, intuitive experience where users can easily explore products, delve into details, and curate a personal list of items in a persistent shopping cart, fostering engagement and paving the way for future transactions.

3. Goals & Success Metrics

Our primary objective for the Simple E-commerce Product Browser is to establish a robust and engaging foundation for product discovery and cart management.

Business Goals:

- **Increase User Engagement:** Foster a positive and intuitive browsing experience that encourages users to explore more products and spend more time on the platform.
- **Validate Core E-commerce Flow:** Prove the viability and user acceptance of the fundamental product browsing and cart population journey before investing in more complex features like checkout and user accounts.
- **Lay Foundation for Future Monetization:** Create a sticky experience that captures user intent, which can be leveraged in future iterations for conversion to sales.

User Goals (Job-to-be-done):

- **Discover Products:** Users want to easily find and explore a variety of products relevant to their interests.
- **Evaluate Products:** Users want to access comprehensive information about a product to make informed decisions.
- **Curate a Shopping List:** Users want to collect and manage a list of desired items without immediate purchase obligation.

Key Performance Indicators (KPIs):

KPI		Target Value Timeframe (Post-Launch)	
Rationale			
• Number of unique product views per session: This measures how deeply users engage with the product catalog. • Target: Average of 5+ unique product views per session. • Timeframe: Within 3 months of launch. • Percentage of products added to cart after viewing details: This indicates the effectiveness of the product detail page in converting interest into action.			

- Target: At least 15% of product detail page views result in an "Add to Cart" action.
- Timeframe: Within 3 months of launch.
- Average number of items in cart per user session: This reflects the user's ability to curate a list of desired items.
- Target: Average of 3+ items in cart per session.
- Timeframe: Within 3 months of launch.
- User retention rate (returning visitors): This measures the stickiness and perceived value of the browsing experience.
- Target: 25% of users returning within 7 days.
- Timeframe: Within 6 months of launch.

4. Target Users & Personas

The Simple E-commerce Product Browser is designed for individuals in the early stages of their shopping journey, primarily focused on discovery and consideration.

Primary Persona:

- Name: Sarah, The Savvy Shopper
- Role: An individual who enjoys browsing online stores for inspiration, comparing products, and building wishlists for future purchases or gifts.
- Goals:
 - Efficiently discover new and interesting products.
 - Gather detailed information about products without feeling rushed.
 - Maintain a curated list of potential purchases in an easily accessible way.
 - Avoid immediate commitment to purchase while exploring options.
- Pain Points:
 - Difficulty in saving items of interest without creating an account or initiating a checkout process.
 - Overwhelming product displays that lack clear information.
 - Slow loading times or clunky interfaces that disrupt the browsing flow.
- Technical Proficiency: Moderate to High. Comfortable with online shopping, expects modern web experiences, uses various devices (desktop, tablet, mobile).

Secondary Persona:

- Name: Mark, The Casual Browser
- Role: Someone who occasionally browses online stores out of curiosity or when

looking for a specific item, but isn't necessarily ready to buy immediately.

- Goals:
- Quickly see what's available in a category.
- Find basic information (price, image) at a glance.
- Easily add an item to a temporary list if it catches their eye.
- Pain Points:
- Being forced to create an account for basic browsing or cart functionality.
- Complex navigation or too many options that distract from simple browsing.
- Unclear pricing or product details.
- Technical Proficiency: Low to Moderate. Prefers simple, intuitive interfaces; may be less patient with complex features or slow performance.

Sarah, The Savvy Shopper, is our primary persona and will drive core design decisions. Her need for detailed product information and a robust, yet flexible, cart experience will ensure the product meets the expectations of engaged online shoppers.

5. Scope

5.1 In Scope

This initial release of the Simple E-commerce Product Browser will focus on delivering a seamless experience for product discovery and cart management. The following core functionalities are explicitly in scope:

- Product Listing Page: A comprehensive display of all available products, showcasing essential information such as product name, price, and a thumbnail image, presented in a clear and scrollable format.
- Product Detail Page: A dedicated page for each product, providing extensive details including the product name, a rich description, current price, and multiple high-resolution images. This page will also feature a prominent "Add to Cart" button.
- Shopping Cart: A persistent, session-based cart where users can collect products of interest. This includes the ability to add items from the product detail page, view a list of added products with their quantities and subtotals, adjust item quantities, and remove items. A visual indicator (e.g., cart icon with item count) will be present across the site.
- Core Functional Requirements:

- Display a list of all available products.
- Allow users to view detailed information for each product.
- Allow users to add a specified quantity of a product to their shopping cart.
- Display the current contents of the user's shopping cart, including product name, quantity, and individual item price.
- Allow users to update the quantity of an item already in the shopping cart.
- Allow users to remove an item from the shopping cart.
- Provide a visual indicator of the number of items in the cart.

5.2 Out of Scope

To maintain focus and deliver a high-quality MVP, the following functionalities are explicitly out of scope for this initial release:

- User Authentication and Profiles: There will be no user login, registration, or personal profile management. Cart contents will be session-based and anonymous.
- Order Placement and Payment Processing: The system will not support checkout, payment gateways, or order finalization. The cart serves purely as a collection mechanism.
- Shipping and Fulfillment Management: No features related to shipping options, delivery tracking, or inventory fulfillment will be included.
- Product Search and Filtering Capabilities: While browsing is central, advanced search bars, category filters, or sorting options are deferred. Users will navigate by scrolling the full product list.
- Admin Panel for Product Management: Product data will be managed via a simple backend interface or static files, not a user-facing CMS.
- Customer Support Features: No live chat, FAQ sections, or contact forms will be part of this release.
- Reviews and Ratings: Functionality for users to submit or view product reviews and ratings is not included.

These items are excluded to ensure a lean, focused MVP that validates the core product browsing and cart experience. They represent potential future enhancements once the foundational experience is proven.

6. Features & Requirements

6.1 Core Features (MVP)

The following features constitute the Minimum Viable Product (MVP) and are critical for the initial launch.

6.1.1 Product Listing Page

- One-line Summary: Displays a catalog of all available products with essential information.
- Detailed Description: This page will serve as the primary entry point for product discovery. It will present a grid or list view of all products, each represented by a thumbnail image, product name, and price. The layout should be clean, responsive, and allow for easy scrolling through the catalog.
- User Story: As a Savvy Shopper, I want to see a comprehensive list of all available products so that I can quickly browse and identify items of interest.
- Acceptance Criteria:
 - All available products are listed on the page.
 - Each product entry clearly shows its name, price, and a representative thumbnail image.
 - Products are displayed in a clear, consistent, and scrollable format.
 - Clicking on a product entry navigates the user to its respective Product Detail Page.
- Priority: Must-have

6.1.2 Product Detail Page

- One-line Summary: Provides comprehensive information for a single product.
- Detailed Description: This page offers an in-depth view of a selected product. It will prominently display the product name, a detailed description, its price, and multiple high-resolution images. A clear and actionable "Add to Cart" button will be present, allowing users to add the product to their shopping cart.
- User Story: As a Savvy Shopper, I want to view detailed information about a specific product so that I can make an informed decision about adding it to my cart.
- Acceptance Criteria:
 - Displays the product name, a detailed description, its price, and multiple images.
 - Includes a clearly visible and functional 'Add to Cart' button.
 - Users can navigate to this page directly from the Product Listing Page.
 - The page content is dynamically loaded based on the selected product.
- Priority: Must-have

6.1.3 Shopping Cart

- One-line Summary: Allows users to collect products they wish to purchase.
- Detailed Description: The shopping cart provides a temporary holding area for products selected by the user. It will display a list of all added products, showing the product name, quantity, and individual item price. Users will be able to adjust the quantity of each item or remove items entirely. A persistent cart icon/indicator will be visible across the site, displaying the current number of items in the cart. Cart contents will be session-based.
- User Story: As a Savvy Shopper, I want to collect and manage a list of desired products so that I can curate my potential purchases without immediate commitment.
- Acceptance Criteria:
 - Users can add products from the Product Detail Page to the cart.
 - The cart displays a list of added products with their names, quantities, and individual item prices.
 - Users can adjust the quantity of any item currently in the cart (e.g., via +/- buttons or an input field).
 - Users can remove individual items from the cart.
 - A cart icon or indicator prominently shows the total number of items currently in the cart, updating in real-time.
 - The cart calculates and displays a subtotal for all items.
- Priority: Must-have

6.2 Post-MVP Features

There are no features explicitly designated as `is_mvp=false` in the current structured analysis. This PRD focuses solely on the MVP scope. Future iterations would naturally include features such as user authentication, advanced search and filtering, product reviews, and eventually, a full checkout and payment processing flow, as outlined in the "Out of Scope" section.

7. User Flows

7.1 Browse Products and Add to Cart

This flow describes the primary user journey for discovering products and populating the shopping cart.

Actor: Shopper

Happy Path:

1. Shopper lands on the homepage/product listing page. The system displays a grid of available products.
2. Shopper scrolls through the list of products. The system loads more products if available (implicit scrolling).
3. Shopper clicks on a product to view its details. The system navigates to the Product Detail Page for the selected product.
4. Shopper reviews product details (description, price, images). The system presents all relevant information clearly.
5. Shopper clicks 'Add to Cart' button. The system processes the request to add the item.
6. System confirms item added to cart. The cart icon/indicator in the header updates to reflect the new item count (e.g., from 0 to 1). A subtle, non-intrusive confirmation message may appear briefly.
7. Shopper can continue browsing (e.g., navigate back to the product listing) or view the cart (e.g., by clicking the cart icon).

Edge Cases:

- Attempting to add an item with quantity 0 or negative:
- Step 5 (Edge Case): Shopper attempts to add a product with an invalid quantity (e.g., 0 or a negative number).
- System Response: The system displays an error message (e.g., "Please enter a valid quantity") and prevents the item from being added to the cart. The cart count remains unchanged.
- Product is out of stock (Assumption: Not explicitly handled in MVP, but a critical edge case for future):
- Step 5 (Edge Case): Shopper clicks 'Add to Cart' for a product that is currently out of stock.
- System Response (Future Consideration): The system would ideally display a message indicating the product is unavailable and disable the 'Add to Cart' button. For this MVP, we assume all listed products are in stock. (Refer to Open Questions for clarification on out-of-stock behavior).

8. Non-Functional Requirements

These requirements define the quality attributes of the system, ensuring it is robust, performant, and user-friendly.

- Performance:
 - Description: Product listing and detail pages should load quickly to provide a smooth user experience.
 - Target Metric: Page Load Time (PLT) for critical pages (Product Listing, Product Detail, Cart View) shall be less than 3 seconds under normal network conditions (broadband, typical device).
- Security:
 - Description: User session data, particularly cart contents, must be protected from unauthorized access and manipulation.
 - Target Metric: The system shall have no critical or high-severity security vulnerabilities identified in penetration tests conducted by an independent third party prior to launch. All data transmission between client and server shall use HTTPS.
- Scalability:
 - Description: The system must be able to handle an increasing number of products and concurrent users without significant performance degradation.
 - Target Metric: The system shall support 1000 concurrent users with an average response time of less than 5 seconds for all API calls and page loads. This is to accommodate growth beyond the initial 100 concurrent users at launch.
- Accessibility:
 - Description: The user interface should be usable by individuals with diverse abilities, adhering to common web accessibility standards.
 - Target Metric (Assumption): The frontend UI shall aim for WCAG 2.1 Level AA compliance for all core browsing and cart functionalities. This includes proper semantic HTML, keyboard navigation support, and sufficient color contrast.
- Reliability:
 - Description: The website should be consistently available and functional, minimizing downtime for users.
 - Target Metric: The system shall maintain a 99.9% uptime for all core services (Product Catalog, Shopping Cart, Frontend UI) as measured over a 30-day period.

9. Assumptions & Dependencies

9.1 Assumptions

- **Product Data Source:** A reliable source of product data (names, descriptions, prices, images, stock quantities) will be provided and maintained. This data will be accessible by the Product Catalog Service.
- **User Environment:** Users will access the website using a modern web browser (e.g., Chrome, Firefox, Safari, Edge latest two versions) and a stable internet connection.
- **Primary Goal Validation:** The primary goal of this MVP is to validate the core user journey of product discovery and cart population. Success will be measured by user engagement with these features, not by immediate sales conversions.
- **Session-based Cart Sufficiency:** Session management for anonymous carts will be sufficient for the initial phase, meaning cart contents are tied to a browser session and not a persistent user account. This implies that if a user clears their cookies or switches browsers, their cart will be lost.
- **Static Product Data Management:** Product data will be managed via a simple backend interface or static files, not requiring a full-fledged Content Management System (CMS) for this MVP.

9.2 Dependencies

- **Product Catalog Service:** The Frontend UI is dependent on the Product Catalog Service for retrieving all product listings and individual product details.
- **Shopping Cart Service:** The Frontend UI is dependent on the Shopping Cart Service for all cart-related operations (add, remove, update quantity, retrieve cart contents). The Shopping Cart Service itself depends on the Product Catalog Service to validate product IDs and retrieve product information for cart items.
- **Infrastructure Team:** Deployment, hosting, and monitoring infrastructure will be provided and managed by an internal infrastructure team.
- **Design Team:** High-fidelity UI/UX designs and assets will be provided by the design team, based on the wireframes and user flows outlined in this document.
- **Suggested Tech Stack:**
 - **Frontend:** React (for building the user interface).
 - **Backend API:** Node.js/Express (for handling API requests and business logic).
 - **Database:** MongoDB or PostgreSQL (for persistent storage of product and cart data).

10. Open Questions & Risks

#	Question / Risk	Owner	Status
1	What is the expected volume of products in the catalog?	PM	Open
2	Are there any specific design guidelines or branding requirements?	PM	Open
3	How will product images be hosted and managed (e.g., CDN, internal storage)?	PM	Open
4	What is the desired behavior for out-of-stock items (e.g., hide, grey out, allow adding to cart with warning)?	PM	Open
5	Risk: Performance degradation with high concurrent users if database isn't optimized.	Engineering	Open
6	Risk: Session-based carts may lead to user frustration if sessions expire too quickly or are lost.	PM	Open

11. Appendix

11.1 Data Entities

- Product: Represents an item available for sale in the e-commerce store.
- Key Attributes:
 - product_id (unique identifier, e.g., UUID)
 - name (string, e.g., "Wireless Bluetooth Headphones")
 - description (text, detailed product information)
 - price (decimal, e.g., 99.99)
 - image_urls (array of strings, URLs to product images)
 - stock_quantity (integer, current available stock)
- Relationships: None directly within this MVP scope.
- CartItem: Represents a single product entry within a user's shopping cart.
- Key Attributes:
 - cart_item_id (unique identifier for the item in the cart, e.g., UUID)
 - product_id (references the product_id from the Product entity)
 - quantity (integer, number of units of this product in the cart)
 - session_id (string, unique identifier for the user's current browsing session, used for anonymous carts)
- Relationships: Many-to-one with Product (via product_id).

11.2 Modules

- **Product Catalog Service:**
 - **Responsibility:** Manages all product information, including details, images, and availability. It acts as the single source of truth for product data.
 - **Exposes:**
 - **Get all products:** Retrieves a list of all available products with essential details.
 - **Get product by ID:** Retrieves comprehensive details for a specific product.
 - **Depends On:** None (directly, but relies on a database for persistence).
- **Shopping Cart Service:**
 - **Responsibility:** Manages user-specific shopping cart contents. It handles adding, removing, and updating items in a user's session-based cart.
 - **Exposes:**
 - **Add item to cart:** Adds a specified product and quantity to the current user's cart.
 - **Remove item from cart:** Deletes a specific item from the cart.
 - **Update item quantity in cart:** Modifies the quantity of an existing item in the cart.
 - **Get cart contents by user/session:** Retrieves all items currently in the user's cart.
 - **Depends On:** Product Catalog Service (to validate product IDs and fetch product details for display in the cart).
- **Frontend UI:**
 - **Responsibility:** Renders the user interface for product browsing and cart management. It consumes data from the backend services and presents it to the user.
 - **Exposes:** None (as it's a client-side application).
 - **Depends On:** Product Catalog Service (for product data) and Shopping Cart Service (for cart operations and data).

11.3 Possible APIs

The following RESTful API endpoints are envisioned to support the functionality described:

- **GET /products**
 - **Description:** Retrieve a list of all available products.
 - **Authentication Required:** False

- Related Module: Product Catalog Service
- GET /products/{id}
- Description: Retrieve details for a specific product.
- Authentication Required: False
- Related Module: Product Catalog Service
- POST /cart/add
- Description: Add a product to the current user's shopping cart.
- Request Body: { "productId": "string", "quantity": "integer" }
- Authentication Required: False (session-based)
- Related Module: Shopping Cart Service
- GET /cart
- Description: Retrieve the current user's shopping cart contents.
- Authentication Required: False (session-based)
- Related Module: Shopping Cart Service
- PUT /cart/update/{item_id}
- Description: Update the quantity of an item in the shopping cart.
- Request Body: { "quantity": "integer" }
- Authentication Required: False (session-based)
- Related Module: Shopping Cart Service
- DELETE /cart/remove/{item_id}
- Description: Remove an item from the shopping cart.
- Authentication Required: False (session-based)
- Related Module: Shopping Cart Service